



MUST-GSL Electronic Medical Record Innovation Challenge

Hackathon Brief

Developing a low-resource, scalable EMR platform for the MUST Teaching Hospital

Challenge at a Glance

Item	Description
Host	The Academy of Medical Sciences, Malawi University of Science and Technology (MUST), in partnership with Global Solutions Limited (GSL) and project development collaborators
Challenge focus	Design and prototype a low-resource, safe, scalable, and specialist-hospital-ready Electronic Medical Record system
Target participants	MUST students working in interdisciplinary teams
Primary use case	Future MUST Teaching Hospital, with phased applicability to campus clinic, district hospitals, diagnostics, specialist clinics, dialysis, public health reporting, teaching, and research
Core output	Functional prototype, system design document, workflow diagrams, governance and safety framework, sustainability plan, and final demo

1. Intellectual Property and Branding Notice

This Electronic Medical Record Innovation Challenge is a joint innovation initiative between the Academy of Medical Sciences (AMS), Malawi University of Science and Technology (MUST), and Global Solutions Limited (GSL). All outputs produced under the challenge, including software code, system designs, user interface designs, data models, workflow diagrams, documentation, presentations, prototype screens, and related materials, shall be governed by the formal participation terms issued by the organizers (Define organizers before this statement so that they are explicitly mentioned in the document). The successful EMR solution selected for further development, deployment, or commercialization shall be treated as jointly held intellectual property of MUST, GSL, and the registered members of the winning development team, with AMS acting as the relevant academic and institutional host within MUST. The winning team may benefit from approved licensing, deployment, maintenance, support, upgrade, or commercialization arrangements, including licensing to the future MUST Teaching Hospital and other approved facilities, but shall not independently license, sell, assign, deploy, or commercialize the EMR outside the approved MUST–GSL–developer framework. All participating teams must include the approved AMS, MUST, and GSL logos on every page of submitted documents, presentations, prototype screens, and interface mock-ups, using the official logo files supplied by the organizers.

2. Call For a Digital Platform Design Competition

The Academy of Medical Sciences, Malawi University of Science and Technology, in partnership with Global Solutions Limited and project development collaborators, is inviting MUST students to participate in an innovation challenge to design and prototype an Electronic Medical Record system that can support the establishment and future operations of the MUST Teaching Hospital.

This challenge is not simply about building software. It is about creating a digital health platform that can improve patient care, strengthen clinical governance, support teaching and research, and demonstrate how locally developed technology can respond to real health system needs in Malawi.

The desired EMR should be:

- Clinically safe and designed around patient safety.
- Easy to learn and use by health workers in busy clinical environments.
- Suitable for low-resource settings with intermittent power, limited bandwidth, and variable infrastructure.
- Scalable from a campus clinic to a teaching hospital and specialist referral facility.
- Capable of future integration with national health information systems and digital health standards.
- Secure, privacy-conscious, and compliant with relevant Malawian legal and governance frameworks.
- Designed with sustainability, local technical capacity, and long-term institutional ownership in mind.

3. Background and Rationale

The AMS at MUST is advancing the establishment of a teaching hospital as part of its broader mission to strengthen medical education, clinical service delivery, applied research, innovation, and specialist health system capacity.

A teaching hospital requires strong clinical workflows, reliable patient records, safe prescribing systems, efficient billing, appropriate reporting tools, dependable diagnostic integration, and a secure data environment for education, quality improvement, and research.

Electronic Medical Record systems are now central to modern health service delivery. However, large commercial systems are often expensive, complex, and difficult to adapt to low-resource settings. For the MUST Teaching Hospital,

the priority is not to replicate large-scale commercial platforms. The priority is to develop a practical, context-aware, modular EMR prototype that can evolve over time.

The World Health Organization Global Strategy on Digital Health emphasizes that digital health initiatives require appropriate strategies, resources, governance, and implementation capacity. [2] Malawi's Digital Health Strategy 2020-2025 and Digital Health Governance Framework 2025-2030 provide relevant national direction on digital health coordination, governance, interoperability, data security, privacy, sustainability, and health system alignment. [3] [4]

This challenge therefore asks students to combine technical creativity with clinical realism, governance awareness, cybersecurity thinking, and sustainable implementation planning.

4. Challenge Objective

The objective of the hackathon is to design and prototype a functional EMR system suitable for phased deployment in:

- The MUST Teaching Hospital.
- MUST campus clinic or outpatient services.
- Dialysis centres and chronic care units in future phases.
- Specialist clinical services and diagnostic departments.

Teams are expected to produce a working prototype, supported by clear documentation, system architecture, workflow diagrams, safety safeguards, governance arrangements, and an implementation plan.

The prototype does not need to be a complete hospital-wide EMR. It should demonstrate a strong minimum viable product that can be expanded through future development phases.

5. Target Participants

The challenge is open to students at the Malawi University of Science and Technology. Interdisciplinary teams and mixed-level/cross-level / mentorship teams are strongly encouraged.

Each team should ideally include members with complementary skills in software development, user interface design, health workflow analysis, cybersecurity, data management, and communication.

6. Problem Statement

Health facilities in Malawi require digital systems that can support safe, efficient, and coordinated care while functioning in environments where resources may be constrained. These constraints may include intermittent power, unstable internet connectivity, limited ICT infrastructure, heavy clinical workloads, limited digital literacy among some users, and the need to align with Ministry of Health reporting requirements.

7. Design Principles

7.1 Patient Safety First

The EMR must reduce clinical risk, not create additional risk. It should support accurate documentation, safe prescribing, allergy alerts, abnormal result notifications, and clear accountability.

7.2 Simplicity and Usability

The system should be easy for nurses, clinicians, pharmacists, laboratory personnel, imaging personnel, billing officers, administrators, students, and ICT staff to learn and use.

7.3 Malawi Context Fit

The system should reflect local clinical workflows, patient identification challenges, village and Traditional Authority structures, connectivity limitations, and the realities of public and private health service delivery in Malawi.

7.4 Scalability

The prototype should be modular and capable of phased growth from an outpatient clinic to a teaching hospital and specialist referral facility.

7.5 Interoperability

The system should be designed with future integration in mind, including compatibility with standards such as DHIS2, HL7 FHIR for health data exchange, DICOM for medical imaging, and LOINC for laboratory observations. [7] [8] [9], and integration with the Malawi Ministry of Health MaHIS, DHIS2 and other systems, through the Health Information Exchange API, with the goal of achieving "One patient, one record".

7.6 Security and Privacy

The system must protect patient data through role-based access, authentication, encryption, logging, audit trails, and appropriate data governance.

7.7 Sustainability

The system must include a realistic plan for maintenance, upgrades, training, local ownership, and long-term technical support.

8. Expanded Core Functional Requirements

Each team must design the EMR as a modular hospital information platform capable of supporting the MUST Teaching Hospital as a specialist, teaching, research, and referral-oriented facility. The system should begin with core patient care workflows, diagnostics, dialysis and hospital management but be designed for future growth into other specialist services, teaching, research and public health reporting.

The prototype needs to fully implement in core modules and demonstrate scale to later modules. Teams must show how the modules will fit together logically and how the system can expand over time.

8.1 Core Modules

8.1.1 Patient Registration, Identification, and Master Patient Index

- Patient registration and demographic profile.
- Unique hospital patient number.
- National ID or other government-issued identifier field, where available.
- Name, sex, date of birth, age, phone number, and address.
- Guardian, parent, next-of-kin, or emergency contact details.
- Village, Traditional Authority, district, and region.
- Occupation, school, workplace, or institution, where relevant.
- Patient category such as outpatient, inpatient, student, staff, private patient, referred patient, emergency patient, or research participant.
- Biometric-ready design, where future implementation is permitted and ethically approved.
- Duplicate patient detection and patient merge workflow.
- Visit and encounter history across services.
- Referral source and referral destination.
- Patient consent status for care, teaching, research, and data use where applicable.

8.1.2 Appointment, Queue, and Patient Flow Management

- Appointment booking and clinic scheduling.
- Walk-in visit registration.
- Provider scheduling.
- Patient check-in and check-out.
- Queue management and triage prioritization.
- Emergency flagging and fast-track workflow.
- Referral to another department.
- Missed appointment tracking.
- Follow-up appointment generation.
- SMS or mobile notification concept for reminders.

8.1.3 Outpatient Clinical Documentation

- Presenting complaint and history of presenting complaint.
- Past medical and surgical history.
- Medication, allergy, social, and family history.
- Review of systems and examination findings.
- Diagnosis and differential diagnosis.
- Clinical plan, orders, prescriptions, referrals, and follow-up plan.
- Structured templates for common clinics.
- Clinician signature, timestamp, and audit trail.

8.1.4 Inpatient and Ward Management

- Admission request and admission diagnosis.
- Ward and bed allocation.
- Transfers between wards or services.
- Ward round notes and progress notes.

- Nursing care plans.
- Fluid balance charts and intake-output monitoring.
- Medication administration record.
- Observation charts.
- Procedure notes.
- Discharge planning and discharge summary.
- Death documentation.
- Inpatient billing linkage.
- Bed occupancy dashboard.

8.1.5 Emergency and Triage Module

- Emergency registration and minimal data rapid registration for unstable patients.
- Triage category and presenting emergency condition.
- Vital signs and emergency alerts.
- Resuscitation notes and trauma notes.
- Time-critical event recording.
- Referral to theatre, ICU, imaging, laboratory, or ward.
- Emergency outcome: discharge, admission, referral, or death.

8.1.6 Nursing Documentation and Care Coordination

- Nursing assessment and nursing problem list.
- Nursing notes and care plans.
- Vital signs monitoring and pain assessment.
- Fall risk and pressure sore risk assessment.
- Wound care documentation.
- Medication administration record.
- Nursing handover.
- Escalation of abnormal findings.
- Patient education and discharge counselling notes.

8.1.7 Vital Signs, Observations, and Clinical Monitoring

- Temperature, blood pressure, pulse rate, respiratory rate, and oxygen saturation.
- Weight, height, body mass index, and pain score.
- Blood glucose and Glasgow Coma Scale, where applicable.
- Early warning score concept.
- Pediatric age-adjusted vital sign alerts.
- Pregnancy status where relevant.
- Abnormal value alerts and trend charts for repeated observations.

8.1.8 Dedicated Physician/Provider Documentation Section

- History & Physical (H&P)
- Admission Notes
- Progress Notes
- Consultation Notes

- Procedure Notes
- Discharge Summaries
- Medication Reconciliation
- Clinical Decision-Making Documentation.

8.1.9 Provider Workflow Integration

- Order entry and management
- Diagnostic result review and acknowledgment
- Interdisciplinary communication tools
- Escalation and critical value notification process
- Care plan updates

8.1.10 Laboratory Information Management

- Laboratory test ordering.
 - Specimen type, sample collection status, and barcode-ready sample labelling.
 - Sample receipt and test processing status.
 - Result entry, verification, and approval.
 - Critical result alerts and abnormal result flagging.
 - Result history and printable laboratory reports.
 - Laboratory workload dashboard and turnaround time tracking.
 - Reagent and consumables inventory concept.
 - Quality control documentation.
 - Linkage to external or referral laboratories.
 - Future mapping to laboratory standards such as LOINC for health measurements, observations, and documents.
- [9]

8.1.11 Pharmacy, Prescribing, and Medication Safety

- Electronic prescribing.
- Drug name, formulation, dose, route, frequency, and duration.
- Allergy alerts and duplicate therapy warning.
- Drug interaction warning concept.
- Pediatric dosing safeguards.
- Pregnancy and breastfeeding warning concept.
- Renal dose adjustment warning concept.
- Prescription approval workflow where required.
- Dispensing status and medication administration linkage.
- Medication history.
- Stock availability indicator.
- Controlled medicines tracking concept.
- Pharmacy workload dashboard.

8.1.12 Dialysis and Chronic Kidney Disease Module

- CKD diagnosis and staging.
- Dialysis registration and dialysis prescription.

- Dialysis session record.
- Pre- and post-dialysis weight.
- Fluid removal target.
- Vascular access type.
- Complications during dialysis.
- Laboratory monitoring and medication tracking.
- Dialysis schedule and missed session tracking.
- Longitudinal chronic care dashboard.

8.1.13 Intensive Care, High-Dependency, and Critical Care Module

- ICU or HDU admission note.
- Continuous observation charting.
- Ventilation status and oxygen therapy documentation.
- Fluid balance and infusion monitoring.
- Inotropes, sedation assessment, and critical care medication documentation.
- Critical care procedure notes.
- Sepsis alert concept.
- Critical result alerts.
- Nursing care plans and daily ICU review notes.
- ICU discharge summary.
- Mortality and morbidity review dashboard.

8.1.14 Billing, Insurance, and Revenue Cycle Management

- Service-based billing.
- Consultation, laboratory, imaging, pharmacy, procedure, theatre, admission, bed, and consumables charges.
- Invoice and receipt generation.
- Payment status and bank or mobile money payment reference.
- Insurance or institutional payer field.
- Waiver or exemption approval workflow.
- Revenue dashboard and unpaid bills report.

8.1.15 Inventory, Supplies, and Biomedical Equipment Linkage

- Pharmacy stock, laboratory reagents, imaging consumables, theatre consumables, and ward supplies.
- Stock alerts and expiry tracking.
- Batch tracking.
- Equipment maintenance records.
- Biomedical equipment downtime reporting.
- Linkage between clinical use and inventory consumption.

8.1.16 Clinical Governance & Patient Safety Components

- Documentation compliance standards
- Authentication and electronic signatures
- Regulatory and accreditation requirements
- Audit trail functionality

8.1.17 Multidisciplinary Coordination

- Nursing-provider communication workflow
- Pharmacy review integration
- Allied health documentation
- Case management and discharge planning collaboration

8.1.18 Health Information Exchange and Interoperability

- API-ready architecture.
- HL7 FHIR-inspired data exchange. [7]
- DICOM-ready imaging integration. [8]
- LOINC-ready laboratory coding. [9]
- ICD-10 or ICD-11 diagnosis coding readiness. [10]
- Ministry of Health reporting system linkage.
- National digital health infrastructure linkage.
- Laboratory equipment interfaces.
- Mobile money interfaces.
- mHealth application integration.
- Secure data export and import.

8.1.19 Administration, Governance, and Audit

- User account management.
- Role-based access control.
- Department configuration.
- Service catalogue and facility configuration.
- Audit trails and access logs.
- Data backup monitoring.
- System usage reports.
- Change control log.
- Incident reporting.
- Downtime reporting.
- Data governance dashboard.

8.2 Future Modules

8.2.1 Digital Medical Imaging and Radiology Module

- Imaging request entry and clinical indication.
- Modality selection such as X-ray, ultrasound, CT, MRI, fluoroscopy, mammography, echocardiography, or interventional imaging.
- Pregnancy status warning for relevant imaging.
- Contrast allergy and renal function warning concept.
- Imaging appointment scheduling and imaging status tracking.
- Radiographer notes.
- Radiologist or specialist report entry.

- Report verification and critical imaging finding alert.
- Image attachment or image-link concept.
- Integration-ready design for Picture Archiving and Communication System and Radiology Information System.
- Teaching image repository concept using de-identified cases.
- Audit trail for image access.
- Imaging workload dashboard and turnaround time tracking.
- Future compatibility with DICOM for medical imaging information. [8]

8.2.2 Theatre, Anaesthesia, and Procedure Management

- Theatre booking and procedure scheduling.
- Pre-operative checklist and consent documentation.
- Anaesthesia assessment.
- Surgical safety checklist.
- Procedure notes and anaesthesia record.
- Implant or device documentation.
- Post-operative orders and recovery room notes.
- Complication documentation.
- Theatre utilization dashboard.
- Procedure coding and billing linkage.

8.2.3 Maternal, Neonatal, and Child Health Module

- Antenatal care record.
- Obstetric history and pregnancy risk assessment.
- Labour and delivery record.
- Partograph concept.
- Caesarean section documentation.
- Postnatal care record.
- Neonatal assessment and birth outcomes.
- Immunization record.
- Pediatric growth monitoring.
- Under-five clinic linkage.
- Maternal and neonatal morbidity and mortality reporting.

8.2.4 Other Specialist Clinic Modules

- Modular templates for internal medicine, surgery, obstetrics and gynaecology, paediatrics, oncology, renal medicine, dialysis, cardiology, neurology, orthopaedics, ophthalmology, ENT, dermatology, psychiatry and mental health, rehabilitation, nutrition, dental services, physiotherapy, and occupational therapy.
- Ability to use general EMR functions while adding specialty-specific templates, forms, scoring tools, and follow-up schedules.

8.2.5 Oncology and Cancer Care Module

- Cancer diagnosis record and cancer staging.
- Histology and pathology linkage.
- Tumour board notes.

- Chemotherapy protocol documentation.
- Radiotherapy referral tracking.
- Surgical oncology notes.
- Treatment cycles and toxicity monitoring.
- Follow-up schedules.
- Palliative care referral.
- Cancer registry reporting concept.
- Multidisciplinary team meeting documentation.

8.2.6 Blood Bank and Transfusion Module

- Blood request and indication for transfusion.
- Blood group and crossmatch status.
- Blood product type and unit number.
- Compatibility check.
- Transfusion start and stop time.
- Transfusion reaction documentation.
- Blood stock dashboard.
- Traceability from request to administration.

8.2.7 Infection Prevention, Antimicrobial Stewardship, and Public Health Surveillance

- Infection diagnosis tracking.
- Notifiable disease alerts.
- Antimicrobial prescribing dashboard.
- Culture and sensitivity linkage.
- Hospital-acquired infection tracking.
- Isolation status.
- TB screening, HIV service linkage, and malaria reporting.
- Outbreak alert concept.
- Antimicrobial stewardship review notes.
- Mortality and morbidity review data.

8.2.8 Rehabilitation, Physiotherapy, and Allied Health Module

- Physiotherapy assessment.
- Occupational therapy assessment.
- Rehabilitation plan.
- Functional status scoring.
- Assistive device documentation.
- Nutrition assessment and dietetic plan.
- Speech and language therapy notes.
- Follow-up scheduling.
- Multidisciplinary team notes.

8.2.9 Teaching, Training, and Simulation Support

- Teaching case flagging.

- De-identified case export.
- Student access mode.
- Supervisor review.
- Simulation patient environment.
- Clinical skills assessment support.
- Audit and feedback dashboards.
- Case discussion notes.
- Access controls separating real patient care from teaching simulations.

8.2.10 Research, Audit, and Quality Improvement Module

- De-identified data extraction.
- Research registry concept.
- Consent tracking.
- Study participant flag where approved.
- Data dictionary.
- Audit reports and quality improvement indicators.
- Morbidity and mortality review reports.
- Ethics-controlled data access workflow.
- Export to statistical software formats.
- Dashboard for clinical indicators.

9. Mandatory Critical System Enhancements

Each team must include the following mandatory enhancements in its design and presentation.

9.1 Clinical Governance Structure

- Medical Director oversight.
- Nursing leadership role.
- Pharmacy oversight.
- Laboratory and diagnostics oversight.
- Health information or records management oversight.
- ICT leadership.
- Clinical informatics leadership.
- Data governance committee.
- Patient safety committee linkage.
- System change control and approval process.
- Clear approval pathway for new forms, alerts, user roles, templates, and system upgrades.

9.2 Patient Safety Framework

- Allergy alerts.
- Drug interaction warning concept.
- Duplicate order prevention.
- Critical laboratory result alerts.
- Critical imaging result alerts.

- Abnormal vital sign triggers.
- Pediatric dosing safeguards.
- Duplicate patient record warning.
- Mandatory fields for high-risk clinical activities.
- Time-stamped clinical notes.
- Escalation alerts for emergency or abnormal findings.
- User accountability through audit logs.
- Alert prioritization to reduce alert fatigue.

9.3 Legal, Ethical, and Compliance Requirements

- Alignment with the Malawi Electronic Transactions and Cyber Security Act, Act No. 33 of 2016. [5]
- Alignment with Malawi's Data Protection Act, 2024. [6]
- Role-based access control.
- Strong authentication.
- Audit trails and time-stamped entries.
- Data minimization and patient confidentiality safeguards.
- Controlled access to sensitive records.
- Secure backup and restoration.
- Privacy-by-design principles.
- HIPAA-informed best practices, without treating HIPAA as the governing law in Malawi.

9.4 Cybersecurity Requirements

- Secure login.
- Multi-factor authentication concept.
- Role-based access control.
- Password policy.
- Session timeout.
- Encryption of data in transit.
- Encryption of sensitive data at rest, where feasible.
- Access logging and failed login tracking.
- Administrator activity logging.
- Backup and recovery plan.
- Incident response plan.
- Secure coding practices.
- Protection against common web vulnerabilities such as SQL injection, cross-site scripting, and insecure direct object references.

9.5 Interoperability and Standards

- Design for Ministry of Health reporting systems.
- Design for laboratory information systems and laboratory equipment interfaces.
- Design for medical imaging systems including PACS and RIS.
- Design for pharmacy inventory and prescribing systems.
- Design for mobile money and patient communication tools.
- Use API-ready architecture and standards-aware data modelling.
- Reference platforms such as OpenMRS may be studied for inspiration; OpenMRS describes itself as using REST and FHIR APIs for integration with other tools and systems. [11]

10. System Resilience for the Malawi Context

The proposed EMR must be designed for low-resource and variable infrastructure environments.

- Offline data entry.
- Synchronization when internet connectivity is restored.
- Local server fallback.
- Cloud or hybrid deployment options.
- Power outage recovery.
- Data backup and restoration testing.
- System uptime monitoring.
- Low-bandwidth optimization.
- Use on standard laptops, desktops, tablets, or mobile devices.
- Simple hardware requirements.
- Local technical support model.
- Clear explanation of how the system will avoid data loss during power interruptions or network failure.

11. Workflow Design Requirements

Each team must include workflow diagrams showing how the EMR supports real clinical operations.

11.1 Patient Journey

- Registration.
- Triage.
- Consultation.
- Orders.
- Laboratory or imaging.
- Diagnosis.
- Treatment.
- Pharmacy.
- Billing.
- Follow-up.
- Referral, admission, discharge, or death documentation.

11.2 Nursing Workflow

- Triage.
- Vital signs.
- Nursing notes.
- Medication administration support.
- Escalation of abnormal findings.
- Ward handover, if inpatient workflow is included.

11.3 Clinician Workflow

- Patient review.
- Documentation.
- Diagnosis.
- Orders.
- Prescribing.
- Review of laboratory and imaging results.
- Discharge or follow-up planning.

11.4 Laboratory Workflow

- Order receipt.
- Sample collection status.
- Result entry.
- Result verification.
- Critical result alert.

11.5 Medical Imaging Workflow

- Imaging request.
- Clinical indication and safety checks.
- Scheduling.
- Image acquisition.
- Radiology reporting.
- Critical finding alert.
- Report review by clinician.

11.6 Pharmacy Workflow

- Prescription receipt.
- Allergy or interaction check.
- Dispensing.
- Stock status.
- Dispensing record.

11.7 Billing Workflow

- Service capture.
- Invoice generation.
- Payment status.
- Receipt reference.
- Revenue reporting.

12. Prototype Development Expectations

Each team must submit a prototype that demonstrates the most important workflows. The prototype may be a web application, mobile-responsive web application, mobile application, hybrid prototype, clickable user interface prototype with partial backend functionality, or functional minimum viable product with database integration.

At minimum, the prototype should demonstrate:

- User login.
- Role-based dashboard.
- Patient registration and patient search.
- Patient profile.
- Clinical encounter documentation.
- Vital signs entry.
- Laboratory order and result entry.
- Medical imaging request and report concept.
- Prescription entry and dispensing status.
- Basic billing and payment status.
- Dashboard or analytics page.
- Audit trail concept.
- Backup, offline synchronization, or local server fallback concept.

Teams should prioritize functionality, clarity, safety, usability, and realistic workflows over visual complexity.

13. Suggested Technology Options

Teams may choose their preferred technology stack, provided they can justify it.

- Frontend: React, Angular, Vue, Flutter, HTML/CSS/JavaScript, or other suitable frameworks.
- Backend: Node.js, Django, Laravel, Spring Boot, .NET, FastAPI, or other suitable frameworks.
- Database: PostgreSQL, MySQL, SQLite, MongoDB, or other appropriate databases.
- APIs: RESTful APIs, FHIR-inspired endpoints, or modular service APIs.
- Deployment: local server, cloud sandbox, containerized deployment, or hybrid design.
- Version control: GitHub, GitLab, or other repository platforms.
- Security: secure authentication, input validation, encrypted connections, access logs, and secure configuration management.

Teams should avoid technologies that are unnecessarily expensive, difficult to maintain locally, or unsuitable for low-resource deployment.

14. Training and Adoption Strategy

Each solution must include a user adoption plan covering:

- Staff orientation.
- Role-specific training.
- Super-user model.
- Nursing champions.
- Clinician champions.
- Pharmacy, laboratory, and imaging champions.
- ICT support.
- Helpdesk or issue reporting system.

- Simulation before live use.
- User feedback collection.
- Continuous improvement process.
- Training materials for new staff.
- Basic digital literacy support.
- Downtime procedure training.

Teams should show how the system will be introduced without disrupting patient care.

15. Financial Sustainability and Maintenance

Each team must include a sustainability model addressing:

- Licensing approach.
- Whether the platform is open-source, proprietary, or hybrid.
- Maintenance costs.
- Hosting costs.
- Technical support.
- Upgrade pathway.
- Bug fixing and security patching.
- Documentation.
- Local developer involvement.
- Student innovation pipeline.
- Potential commercialization pathway.
- Long-term governance by MUST and GSL.

The strongest submissions will show how the EMR can be maintained locally and improved over time without creating unsustainable vendor dependence.

16. Expanded Future Development Roadmap

The EMR should be presented as a phased platform that can grow with the MUST Teaching Hospital.

Phase 1: Core Outpatient and Campus Clinic EMR

- Patient registration.
- Patient search.
- Visit history.
- Clinical notes.
- Vital signs.
- Diagnosis and treatment plan.
- Basic laboratory orders.
- Basic prescriptions.
- Basic billing.
- User login and role-based access.

Phase 2: Diagnostic Services Integration

- Expanded laboratory module.

- Sample tracking.
- Critical result alerts.
- Imaging requests.
- Radiology reporting.
- PACS/RIS integration concept.
- DICOM-ready imaging workflow.
- Laboratory terminology mapping.
- Diagnostic turnaround time dashboards.

Phase 3: Inpatient, Ward, and Nursing Care

- Admissions.
- Bed management.
- Ward rounds.
- Nursing care plans.
- Medication administration record.
- Fluid balance.
- Inpatient progress notes.
- Discharge summaries.
- Mortality tracking.

Phase 4: Pharmacy, Inventory, and Medication Safety

- Advanced prescribing.
- Drug interaction checking.
- Allergy and contraindication alerts.
- Stock management.
- Expiry tracking.
- Batch tracking.
- Controlled medicines tracking.
- Pharmacy analytics.

Phase 6: Specialist Clinical Services

- Oncology.
- Dialysis and renal care.
- Cardiology.
- Paediatrics.
- Obstetrics and gynaecology.
- Mental health.
- Rehabilitation.
- Nutrition.
- Palliative care.
- Chronic disease clinics.

Phase 5: Theatre, Anaesthesia, and Critical Care

- Theatre booking.

- Surgical checklist.
- Anaesthesia assessment.
- Procedure notes.
- Recovery room notes.
- ICU and HDU charting.
- Ventilation and infusion records.
- Critical care dashboards.

Phase 7: Maternal, Neonatal, and Child Health

- Antenatal care.
- Labour and delivery record.
- Partograph concept.
- Postnatal care.
- Neonatal record.
- Immunization.
- Pediatric growth monitoring.
- Maternal and neonatal outcomes reporting.

Phase 8: Public Health and Ministry Reporting

- HIV, TB, malaria, and non-communicable disease indicators.
- Notifiable disease alerts.
- Vaccination dashboards.
- Disease surveillance.
- Facility performance dashboards.
- Ministry reporting exports.
- Outbreak monitoring concept.

Phase 9: Teaching, Research, and Quality Improvement

- De-identified teaching case repository.
- Student learning environment.
- Clinical audit reports.
- Research data extraction.
- Ethics-controlled data access.
- Quality improvement dashboards.
- Morbidity and mortality review support.

Phase 10: Advanced Digital Health and Innovation

- Patient portal.
- SMS reminders.
- Telemedicine integration.
- Mobile health applications.
- AI-assisted decision support, subject to governance and validation.
- Predictive analytics.
- Remote monitoring.

- Wearable device integration.
- Regional referral network integration.

17. Hackathon Format

17.1 Team Formation

- Teams register their team name, members, proposed solution title, and selected technology approach.

17.2 Ideation and Workflow Mapping

- Teams define user groups, clinical workflows, patient journey, system architecture, and minimum viable prototype scope.

17.3 Mentored Build Sprint

- Teams develop prototypes with support from mentors in clinical workflows, software engineering, data management, cybersecurity, health informatics, business sustainability, and user interface design.

17.5 Prototype Testing

- Teams test their prototypes using synthetic patient scenarios.

17.6 Final Demo and Judging

- Teams present their prototype, design rationale, safety framework, sustainability model, and implementation plan to a judging panel.

18. Indicative Timeline

Timelines are indicative from the Week beginning 1st June 2026 (Week 0) as follows:

Stage	Activity	Indicative Timing
Stage 1	Launch of challenge and publication of brief	Week 0
Stage 2	Team registration and orientation	Week 1
Stage 3	Ideation, workflow mapping, and design submission	Week 2
Stage 4	Prototype development sprint	Weeks 3-5
Stage 5	Mentorship clinics and interim review	Week 4
Stage 6	Final prototype submission	Week 6
Stage 7	Demo day and judging	Week 6 - 7
Stage 8	Awards, feedback, and next-stage incubation	After final judging

19. Expected Deliverables

19.1 System Design Document

- Problem definition.
- Target users.
- System modules.
- User roles.
- System architecture.
- Data model.
- Security design.
- Deployment model.
- Assumptions and limitations.

19.2 Functional Prototype

- A web, mobile, or hybrid prototype demonstrating the key workflows.

19.3 Workflow Diagrams

- Patient journey.
- Nursing workflow.
- Clinician workflow.
- Laboratory workflow.
- Medical imaging workflow.
- Pharmacy workflow.
- Billing workflow.

19.4 Governance and Patient Safety Framework

- Clinical governance structure.
- Data governance structure.
- Patient safety alerts.
- Change control process.
- Audit trail approach.
- Accountability model.

19.5 Cybersecurity and Compliance Plan

- Authentication.
- Access control.
- Audit logging.
- Data protection.
- Incident response.
- Backup and recovery.
- Legal and ethical considerations.

19.6 Sustainability and Implementation Plan

- Maintenance model.
- Hosting approach.
- Cost considerations.
- Training plan.
- Support model.
- Upgrade pathway.
- Scale-up plan.

19.7 Final Presentation and Demo

- Problem summary.
- Proposed solution.
- Prototype demonstration.
- Innovation features.
- Clinical relevance.
- Malawi context fit.
- Patient safety features.
- Sustainability model.
- Future development roadmap.

20. Judging Criteria

Category	Weight	What Judges Will Look For
Clinical Relevance	20%	Does the system respond to real clinical workflows and teaching hospital needs?
Patient Safety	20%	Does the system reduce clinical risk through alerts, documentation, audit trails, and safe workflows?
Innovation	15%	Does the solution show creativity, originality, and practical problem-solving?
Technical Design	15%	Is the architecture logical, scalable, secure, and maintainable?
Malawi Context Fit	15%	Does the design address low-resource realities, local patient identification, offline use, and future national integration?
Sustainability	15%	Is there a realistic plan for maintenance, training, upgrades, and long-term local ownership?

Total score: 100%.

21. Prize Awards

The competition will award a total prize fund of MK 1,000,000, distributed among the top three winning teams as follows:

Position	Prize Amount
First Place	MK 3,000,000.00
Second Place	MK 2,000,000.00
Third Place	MK 1,000,000.00

22. Bonus Recognition Areas

Judges may give additional recognition to teams that demonstrate:

- Strong user interface design.
- Offline-first architecture.
- Strong accessibility features.
- FHIR-aware data model.
- DICOM-aware imaging workflow.
- LOINC-ready laboratory design.
- Advanced clinical decision support concept.
- Safe pediatric workflows.
- De-identified research data export.
- Strong data visualization.
- Clear implementation roadmap.
- Excellent teamwork and presentation.
- Practical cost-conscious design.

23. Intellectual Property Ownership, Protection and Commercial Licensing

The winning EMR solution shall be managed as a jointly developed innovation arising from a MUST-supported research, consultancy, and technology-development activity undertaken in partnership with Global Solutions Limited and the successful student development team. Accordingly, MUST, GSL, and the registered developers of the successful EMR shall be recognized as joint intellectual property holders in respect of the software code, system architecture, user interface designs, data models, technical documentation, workflow diagrams, prototypes, and related outputs materially developed under the challenge. This arrangement shall be formally documented through an IP and commercialization agreement that records each party's contribution, ownership interest, rights, responsibilities, and benefit-sharing entitlements, consistent with the MUST Research and Consultancy Policy's emphasis on innovation development, technology transfer, commercialization, and equitable benefit sharing for faculty, student researchers, and collaborators.

The successful EMR development team shall be eligible to benefit from approved commercial licensing, deployment, maintenance, support, upgrade, or service arrangements involving the EMR, including licensing or deployment to the MUST Teaching Hospital and any subsequent licensing to other hospitals, clinics, public health institutions, private providers, or partner facilities. Any such benefits shall be administered through the approved MUST–GSL–developer commercialization framework and distributed in accordance with the agreed licensing and benefit-sharing arrangements. This shall allow the developers to be fairly rewarded for their contribution while ensuring that the EMR remains institutionally protected, properly governed, and sustainably commercialized.

For avoidance of doubt, the EMR development team shall not independently sell, assign, license, sublicense, disclose, transfer, commercialize, or deploy the EMR to any external facility or third party outside the approved MUST–GSL–

developer framework. All external licensing, including licensing to facilities other than the MUST Teaching Hospital, shall require prior written authorization through the joint IP and commercialization agreement. This restriction is intended to protect the joint IP, avoid unauthorized use of University-supported outputs, maintain compliance with MUST's research and consultancy governance requirements, and ensure that all commercialization proceeds are transparently managed for the benefit of the recognized joint IP holders.

24. Submission Rules

All submissions must:

- Be produced by registered MUST student teams.
- Include both MUST and GSL logos on every page and prototype screen.
- Use only synthetic or fictional patient data.
- Include all required deliverables.
- Be submitted by the official deadline.
- Acknowledge the joint intellectual property position of MUST and GSL.
- Avoid use of third-party proprietary software, code, data, or images without proper permission.
- Clearly declare any open-source tools, libraries, templates, or frameworks used.
- Include a short statement of limitations and assumptions.

25. What a Strong Submission Looks Like

A strong submission will demonstrate:

- A clear understanding of hospital workflows.
- A simple and intuitive user experience.
- Strong clinical safety thinking.
- A realistic technical architecture.
- Sensitivity to Malawi's infrastructure realities.
- A clear path from prototype to implementation.
- Strong documentation.
- A feasible maintenance and sustainability model.
- Evidence that the team tested the prototype using realistic patient scenarios.
- A convincing explanation of how the system can support teaching, service delivery, research, and quality improvement.

26. Final Message to Students

This challenge gives MUST students an opportunity to contribute directly to a major institutional and national health innovation agenda.

The MUST Teaching Hospital will require digital systems that are safe, efficient, locally relevant, and capable of growth. Through this challenge, students will have the opportunity to apply their skills to a real-world problem with direct relevance to patient care, health worker productivity, clinical teaching, research, and health system strengthening.

The best solutions will not be the most complicated. They will be the most thoughtful, usable, safe, scalable, and realistic.

This is an opportunity to build more than a prototype. It is an opportunity to help shape the digital foundation of a future teaching hospital and contribute to the transformation of healthcare delivery in Malawi.

27. Selected References and Standards

Ref.	Source	Title / Description	Link
[2]	World Health Organization	Global Strategy on Digital Health 2020-2025; published by WHO.	Open source
[3]	Ministry of Health and Population, Malawi	Malawi Digital Health Strategy 2020-2025.	Open source
[4]	Ministry of Health, Malawi	Digital Health Governance Framework for Malawi 2025-2030.	Open source
[5]	Republic of Malawi / MalawiLII	Electronic Transactions and Cyber Security Act, Act No. 33 of 2016.	Open source
[6]	Malawi Data Protection Authority	Data Protection Act, 2024.	Open source
[7]	HL7 International	FHIR Overview: standard for exchanging healthcare information electronically.	Open source
[8]	DICOM Standards Committee / NEMA	DICOM Standard: international standard for medical imaging information.	Open source
[9]	LOINC / Regenstrief Institute	LOINC: international standard for identifying health measurements, observations, and documents.	Open source
[10]	World Health Organization	ICD-11: International Classification of Diseases 11th Revision.	Open source
[11]	OpenMRS	OpenMRS official platform information: open-source EMR with REST and FHIR APIs.	Open source